

Special Functions

S.1

Solution:

Using the definition of $\Gamma(z)$:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

it can be shown that $\Gamma(z+1) = z\Gamma(z)$:

$$\begin{aligned} \Gamma(z+1) &= \int_0^{\infty} t^z e^{-t} dt = \underbrace{[-t^z e^{-t}]_0^{\infty}}_{=0} + \int_0^{\infty} z t^{z-1} e^{-t} dt = \\ &= z \int_0^{\infty} t^{z-1} e^{-t} dt = z\Gamma(z) \end{aligned}$$

We also have

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt = [-e^{-t}]_0^{\infty} = 1$$

This gives

$$\Gamma(3) = \Gamma(2+1) = 2\Gamma(2) = 2\Gamma(1+1) = 2 \cdot 1 \cdot \Gamma(1) \rightarrow \Gamma(3) = 2$$

Next

$$\begin{aligned} \Gamma\left(\frac{1}{2}\right) &= \int_0^{\infty} t^{-\frac{1}{2}} e^{-t} dt \rightarrow [t = x^2 \rightarrow dt = 2x dx] = 2 \int_0^{\infty} x x^{-1} e^{-x^2} dx = \\ &= 2 \int_0^{\infty} e^{-x^2} dx = 2 \frac{\sqrt{\pi}}{2} \rightarrow \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \end{aligned}$$

Finally we will use the definition for $\Gamma(z)$, $\text{Re } z \leq 0$:

$$\Gamma(z) = \frac{\Gamma(z+n)}{(z+n-1)\dots(z+1)z}, \quad n \text{ chosen such that } \operatorname{Re}(z+n) > 0$$

This gives

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}} \rightarrow \Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}$$

□

S.5

Solve the Abelian integral

$$\int_0^x \frac{u(y)}{\sqrt{x-y}} dy = 1, \quad x > 0$$

Solution:

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = F(s)$$

↓

$$\frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} U(s) = \frac{1}{s}$$

↓

$$U(s) = \frac{1}{\Gamma\left(\frac{1}{2}\right)s^{\frac{1}{2}}} = \frac{1}{\Gamma\left(\frac{1}{2}\right)^2} \frac{\Gamma\left(\frac{1}{2}\right)}{s^{\frac{1}{2}}} \xrightarrow{\mathcal{L}^{-1}} u(x)\theta(x) = \frac{1}{\pi} x^{-\frac{1}{2}}\theta(x)$$

↓

$$u(x) = \frac{1}{\pi\sqrt{x}}, \quad x > 0$$

□

S.6

Solution:

□

S.7

Show that

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

Solution:

Using the Fourier expansion

$$e^{ir \sin \theta} = \sum_{-\infty}^{\infty} J_n(r) e^{in\theta}$$

and Parseval's Theorem

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |e^{ir \sin \theta}|^2 d\theta = \sum_{-\infty}^{\infty} |J_n(r)|^2$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \overline{e^{ir \sin \theta}} e^{ir \sin \theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta = \frac{1}{2\pi} 2\pi = 1$$

↓

$$\sum_{-\infty}^{\infty} |J_n(r)|^2 = 1$$

□

S.11

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}$$

(b) Use (a) and

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

to calculate $J_{\frac{3}{2}}(x)$ (c) Show that $J_0'(x) = -J_1(x)$

Solution:

(a) The power series is

$$J_\nu(x) = \left(\frac{x}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k$$

$$\downarrow$$

$$\frac{J_\nu(x)}{x^\nu} = \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^k$$

This can be differentiated termwise as

$$\begin{aligned} \frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) &= \left(\frac{1}{2}\right)^\nu \sum_{k=0}^{\infty} \frac{-2xk}{4k! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{x}{2} \left(\frac{1}{2}\right)^\nu \sum_{\substack{k=1 \\ *}}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} = \\ &= -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=1}^{\infty} \frac{1}{(k-1)! \Gamma(\nu + k + 1)} \left(-\frac{x^2}{4}\right)^{k-1} \rightarrow [k \rightarrow k+1] \rightarrow \\ &\rightarrow -\frac{1}{x^\nu} \left(\frac{x}{2}\right)^{\nu+1} \sum_{k=0}^{\infty} \frac{1}{k! \Gamma(\nu + 1 + k + 1)} \left(-\frac{x^2}{4}\right)^k = \boxed{-\frac{J_{\nu+1}(x)}{x^\nu}} \end{aligned}$$

* (The $k=0$ contribution is zero, since it is a constant and was differentiated.)

(b) Setting $\nu = \frac{1}{2}$ we get

$$\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = -\frac{J_{\frac{3}{2}}(x)}{x^{\frac{1}{2}}}$$

$$\frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\frac{\sqrt{\frac{2}{\pi x}} \sin x}{x^{\frac{1}{2}}} \right) = \frac{d}{dx} \left(\sqrt{\frac{2}{\pi}} \frac{\sin x}{x} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{x \cos(x) - \sin(x)}{x^2} \right)$$

$$\downarrow$$

$$J_{\frac{3}{2}}(x) = -x^{\frac{1}{2}} \frac{d}{dx} \left(\frac{J_{\frac{1}{2}}(x)}{x^{\frac{1}{2}}} \right) = x^{\frac{1}{2}} \sqrt{\frac{2}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^2} \right) = \sqrt{\frac{2x}{\pi}} \left(\frac{\sin x - x \cos x}{x^2} \right)$$

(c) Using the result from (a)

$$\frac{d}{dx} \left(\frac{J_0(x)}{x^0} \right) = -\frac{J_1(x)}{x^0} \rightarrow \boxed{J'_0(x) = -J_1(x)}$$

□

S.12**Solution:**

□

S.13

(a) $r^2 y'' + r y' = 0$

(b) $r^2 y'' + r y' - y = 0$

(c) $r^2 y'' + r y' + r^2 y = 0$

Solution:

Using a definition that the general solution to Bessel's differential equations

$$u'' + \frac{1}{r}u' + \left(\lambda - \frac{\nu^2}{r^2}\right)u = 0$$

are

$$\begin{aligned} & aJ_\nu(\sqrt{\lambda}r) + bY_\nu(\sqrt{\lambda}r), & \text{if } \lambda > 0 \\ & ar^\nu + br^{-\nu}, & \text{if } \lambda = 0, \nu \neq 0 \\ & a + b \ln r, & \text{if } \lambda = \nu = 0 \end{aligned}$$

(a)

$$r^2 y'' + r y' = 0 \rightarrow y'' + \frac{1}{r}y' = 0, \quad \lambda = \nu = 0$$

$$y(r) = a + b \ln r$$

(b)

$$r^2 y'' + r y' - y = 0 \rightarrow y'' + \frac{1}{r}y' - \frac{1}{r^2}y = 0, \quad \lambda = 0, \nu = 1$$

$$y(r) = ar + \frac{b}{r}$$

(c)

$$r^2 y'' + r y' + r^2 y = 0 \rightarrow y'' + \frac{1}{r} y' + y = 0, \quad \lambda = 1, \nu = 0$$

$$y(r) = a J_0(r) + b Y_0(r)$$

□

S.14

$$\mathcal{A}u = -\frac{1}{x}(xu')'$$

$$D_{\mathcal{A}} = \{u \in \mathcal{C}^2(I) \mid u \text{ bounded near } 0, u'(2) = 0\}$$

- (a) Determine a scalar product in which $\mathcal{A}$ is symmetric and positive semidefinite
- (b) Determine the eigenvalues and eigenvectors of $\mathcal{A}$.
- (c) Determine numerically the three lowest eigenvalues.
- (d) Let $u(x) = x$. Write down u 's expansion in eigenfunctions to $\mathcal{A}$. The coefficients should be given in integral form.

Solution:

(a)

$$(u \mid \mathcal{A}v) = \int_0^2 \overline{u(x)} \mathcal{A}v(x) w(x) dx$$

with $w(x) = x$

$$\begin{aligned} (u \mid \mathcal{A}v) &= - \int_0^2 \overline{u} (xv')' dx = - \underbrace{[\overline{u}(xv')]_0^2}_{=0} + \int_0^2 (\overline{u'}x) v' dx = \\ &= \underbrace{\int_0^2 (x\overline{u'}) v' dx}_{= (*)} = \underbrace{[(x\overline{u'})v]_0^2}_{=0} - \int_0^2 (x\overline{u'})' v dx = (\mathcal{A}u \mid v) \end{aligned}$$

To show that it's positive semidefinite, start from (*) and set $u = v$:

$$\int_0^2 x \overline{u'} u' dx = \int_0^2 x |u'|^2 dx \geq 0$$

(b) Setting

$$\mathcal{A}u = \lambda u \rightarrow -\frac{1}{x}(xu')' = -\frac{1}{x}(u' + xu'') = \lambda u$$

$$\downarrow$$

$$-\frac{1}{x}xu'' - \frac{1}{x}u' - \lambda u = 0 \rightarrow u'' + \frac{1}{x}u' + \lambda u = 0$$

which has the solutions (according to the definition of the Bessel's differential equation solutions):

$$aJ_0(\sqrt{\lambda}x) + bY_0(\sqrt{\lambda}x), \quad \lambda > 0, \nu = 0$$

Since Y_0 is unbounded near $x = 0$, b has to be zero to fulfill the requirement of u being bounded near zero.

$$u(x) = aJ_0(\sqrt{\lambda}x)$$

$$u'(2) = \sqrt{\lambda}aJ_0'(\sqrt{\lambda}2) = 0$$

This gives $\lambda_0 = 0$. The rest of the eigenvalues are given by

$$J_0'(2\sqrt{\lambda_k}) = 0 \rightarrow \lambda_k = \left(\frac{\beta_{0,k}}{2}\right)^2$$

The eigenfunctions are then given by

$$\varphi_{k(x)} = J_0\left(\frac{\beta_{0,k}}{2}x\right)$$

(c)

$$\lambda_0 = 0$$

$$\lambda_1 = \left(\frac{3.8317}{2}\right)^2 = 3.67$$

$$\lambda_2 = \left(\frac{7.0156}{2}\right)^2 = 12.31$$

□

S.15

Determine a non-trivial solution to the eigenvalue problem

$$\begin{aligned} -\Delta u &= \lambda u, \text{ in } \Omega \\ u &= 0, \quad \text{on } \partial\Omega \end{aligned}$$

where Ω is the circle sector $0 < r < 1, 0 < \theta < \frac{\pi}{3}$

Solution:

We have the conditions for $u(r, \theta)$

$$\begin{aligned} u(r, 0) &= u\left(r, \frac{\pi}{3}\right) = 0 \\ u(1, \theta) &= 0 \end{aligned}$$

The Laplacian in cylindrical coordinates is

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}$$

Expressing $u(r, \theta)$ with separation of variables gives

$$u(r, \theta) = R(r)\Theta(\theta)$$

Putting this into the Laplacian gives

$$\begin{aligned} R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta'' &= -\lambda R\Theta \\ \downarrow \\ \frac{R''}{R} + \frac{1}{r}\frac{R'}{R} + \frac{1}{r^2}\frac{\Theta''}{\Theta} &= -\lambda \\ \downarrow \\ r^2\frac{R''}{R} + r\frac{R'}{R} + \lambda r^2 &= -\frac{\Theta''}{\Theta} \\ \downarrow \\ \frac{r^2}{R}\left(R'' + \frac{1}{r}R' + \lambda R\right) &= -\frac{\Theta''}{\Theta} \\ \downarrow \\ \frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) &= -\frac{\Theta''}{\Theta} \end{aligned}$$

LHS depend only on r while RHS depend only on θ , therefore both must be a constant

$$\begin{aligned} \frac{r^2}{R}\left(\frac{1}{r}(rR')' + \lambda R\right) &= \beta \rightarrow \frac{1}{r}(rR')' + \lambda R = \frac{\beta R}{r^2} \\ -\frac{\Theta''}{\Theta} &= \beta \\ \downarrow \\ \begin{cases} \Theta'' = -\beta\Theta \\ R'' + \frac{1}{r}R' + R\left(\lambda - \frac{\beta}{r^2}\right) = 0 \end{cases} \end{aligned}$$

Solution to Θ :

$$\Theta(\theta) = A \cos(\sqrt{\beta}\theta) + B \sin(\sqrt{\beta}\theta)$$

$$\Theta(0) = A \rightarrow A = 0$$

$$\Theta\left(\frac{\pi}{3}\right) = B \sin\left(\sqrt{\beta}\frac{\pi}{3}\right) = 0 \rightarrow \sqrt{\beta}\frac{\pi}{3} = k\pi \rightarrow \beta = 9k^2, \quad k = 1, 2, 3\ldots$$

$$\Theta(\theta) = \sin(\sqrt{\beta}\theta)$$

Put the β into the equation for $R(r)$ and we get Bessel's differential equation:

$$R'' + \frac{1}{r}R' + \left(\lambda - \frac{9k^2}{r^2}\right)R = 0, \nu^2 = 9k^2 \rightarrow \nu = 3k$$

This has the solution

$$R(r) = a_k J_{3k}(\sqrt{\lambda}r) + b_k Y_{3k}(\sqrt{\lambda}r)$$

Since $R(r)$ has to be bounded, $b_k = 0$

$$R(r) = a_k J_{3k}(\sqrt{\lambda_k}r)$$

$$R(1) = 0 \rightarrow \lambda_k = \alpha_{3k,n}^2$$

The full solution is then

$$u(r, \theta) = a_k J_{3k}(a_{3k,n}r) \sin(3k\theta)$$

$$\lambda_{kn} = \alpha_{3k,n}^2, \quad k, n = 1, 2, 3\ldots$$

□

S.16

The spherical Bessel functions are defined as

$$j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}}(x)$$

Calculate j_0, j_1 , and j_2 by using

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin(x)$$

$$\frac{d}{dx} \left(\frac{J_\nu(x)}{x^\nu} \right) = -\frac{J_{\nu+1}(x)}{x^\nu}, \quad \nu = 0, 1, 2\ldots$$

Solution:

$$j_0(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{1}{2}}(x) = \sin(x)$$

$$j_1(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{3}{2}}(x) = \frac{\sin(x) - x \cos(x)}{x^2}$$

$$j_2(x) = \sqrt{\frac{\pi}{2x}} J_{\frac{5}{2}}(x) = \sqrt{\frac{\pi}{2x}} (-x^{\frac{3}{2}}) \frac{d}{dx} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right)$$

$$\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} = \frac{1}{x^{\frac{3}{2}}} \sqrt{\frac{2x}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^2} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{\sin(x) - x \cos(x)}{x^3} \right)$$

$$\frac{d}{dx} \left(\frac{J_{\frac{3}{2}}(x)}{x^{\frac{3}{2}}} \right) = \sqrt{\frac{2}{\pi}} \left(\frac{x^4 \sin(x) - 3x^2 \sin(x) + 3x^3 \cos(x)}{x^6} \right)$$

$$j_2(x) = \left(\frac{(3 - x^2) \sin(x) - 3x \cos(x)}{x^3} \right)$$

□

S.18

Determine the general solution to

(a) $r^2 y'' + 2ry' = 0$

(b) $r^2 y'' + 2ry' - 2y = 0$

(c) $r^2 y'' + 2ry' + r^2 y = 0$

Solution:

The general solution to the operator

$$\mathcal{A} = y'' + \frac{2}{r} y' + \left(\lambda - \frac{l(l + \frac{1}{2})}{r^2} \right) y$$

is

$$\begin{cases} aj_l(\sqrt{\lambda}r) + by_l(\sqrt{\lambda}r), & \lambda > 0 \\ ar^l + br^{-l-1}, & \lambda = 0, l \neq -\frac{1}{2} \\ \frac{a+b\ln(r)}{\sqrt{r}}, & \lambda = 0, l = -\frac{1}{2} \end{cases}$$

(a)

$$r^2 y'' + 2ry' = 0 \rightarrow y'' + \frac{2}{r}y' = 0$$

Here we have $\lambda = 0, l = 0$, so the solution is:

$$y(r) = a + \frac{b}{r}$$

(b)

$$r^2 y'' + 2ry' - 2y = 0 \rightarrow y'' + \frac{2}{r}y' - \frac{2}{r^2}y = 0$$

$$\downarrow$$

$$y'' + \frac{2}{r}y' + \left(0 - \frac{2}{r^2}\right)y = 0$$

Here we have $\lambda = 0, l(l+1) = 2 \rightarrow l = 1$:

$$y(r) = ar + \frac{b}{r^2}$$

(c)

$$r^2 y'' + 2ry' + r^2 y = 0 \rightarrow y'' + \frac{2}{r}y' + y = 0$$

$$\downarrow$$

$$y'' + \frac{2}{r}y' + \left(1 - \frac{0 \cdot (0+1)}{r^2}\right)y = 0$$

Here we have $\lambda > 0$ and $l = 0$ (or $l = -1$)

$$y(r) = aj_0(r) + by_0(r)$$

□

S.20

Let $\Omega =$ unit sphere in $\mathbb{R}^3$,

$$\mathcal{A} = -\Delta$$

$$D_{\mathcal{A}} = \{u \in C^2(\Omega) \mid u = 0 \text{ on } \partial\Omega\}.$$

(a) Determine the radial (θ, ϕ independent) eigenfunctions and eigenvalues.

(b) Expand the function $1(x, y, z) \equiv 1$ using eigenfunctions to $\mathcal{A}$.

Solution:

The radial component of the Laplacian in spherical coordinates is

$$\Delta_r = u'' + \frac{2}{r}u'$$

So we want to solve

$$\mathcal{A}u = \lambda u \rightarrow -\Delta_r u = \lambda u$$

$$\downarrow$$

$$u'' + \frac{2}{r}u' + \lambda u = 0$$

$$\downarrow$$

$$u'' + \frac{2}{r}u' + \left(\lambda - \frac{l(l+1)}{r^2}\right)u = 0$$

Here we have $\lambda > 0$ and $l = 0$ and we use the general solution to this and get

$$u(r) = aj_0(\sqrt{\lambda}r) + by_0(\sqrt{\lambda}r)$$

Since $u(r)$ has to be bounded at $r = 0$, the $y_0(\sqrt{\lambda}r)$ -term has to disappear, so $b = 0$ and our final solution is

$$u(r) = aj_0(\sqrt{\lambda}r)$$

The eigenvalues can be found from

$$u(1) = 0 \rightarrow aj_0(\sqrt{\lambda}) = 0$$

which means $\sqrt{\lambda}$ has to make $j_0 = 0$. But $j_0(\sqrt{\lambda}r) = \frac{\sin(\sqrt{\lambda}r)}{r}$ so that gives

$$\sqrt{\lambda} = k\pi$$

$$\downarrow$$

$$\lambda_k = k^2\pi^2$$

and the eigenfunctions are

$$\varphi_k(r) = \frac{\sin(k\pi r)}{r}$$

(b)

$$1 = \sum_{k=1}^{\infty} c_k \varphi_k(r)$$

$$\begin{aligned}
c_k &= \frac{(\varphi_k \mid 1)}{(\varphi_k \mid \varphi_k)} = \frac{\int_0^1 r \sin(k\pi r) \, dr}{\int_0^1 \sin^2(k\pi r) \, dr} = \frac{\left[\frac{-r \cos(k\pi r)}{k\pi} \right]_0^1 + \underbrace{\int_0^1 \frac{\cos(k\pi r)}{k\pi} \, dr}_{=0}}{\int_0^1 \left(\frac{1}{2} - \underbrace{\frac{\cos(2k\pi r)}{2}}_{\int=0} \right) \, dr} = \\
&= -2 \frac{\cos(k\pi)}{k\pi} \rightarrow c_k = \frac{2}{k\pi} (-1)^{k+1}
\end{aligned}$$

□